

South Sudan

conflict map

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An interactive map project commissioned by



WATCH LIST ON CHILDREN AND ARMED CONFLICT

In context

- Borders: Uganda, Kenya, Ethiopia, Sudan, Central African Republic, DR Congo
 - Dinka: 35%
 - Nuer: 15%
- Relevant dates:
 - 1955 Sudan independence
 - 2005 Peace agreement
 - **July 9, 2011 Independence**
 - 2013 Civil war



South Sudan



An interactive, visual tool for Watchlist

Central use is to provide international decision makers (UN member states, UN staff, and policymakers) with information about the conflict and the areas affected by it.

The Project



How are medical facilities and access to humanitarian aid affected by the conflict in South Sudan?

Research question

Step 1

Reports → Points

Data Sources:

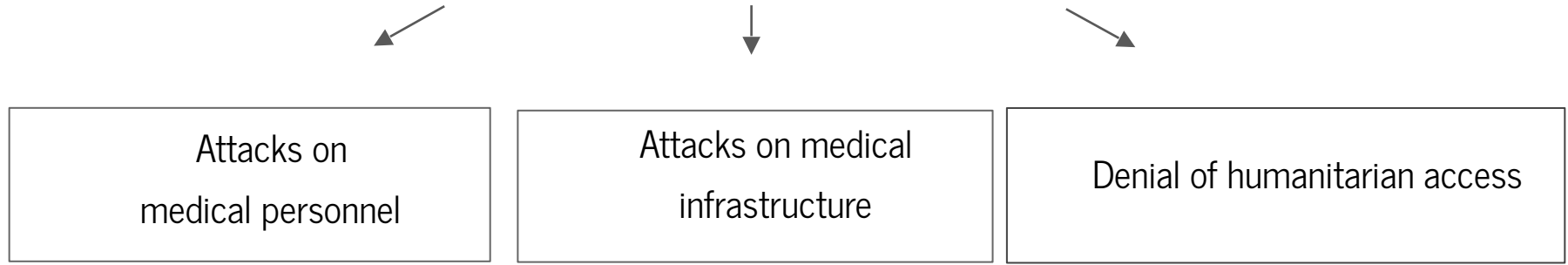
- ▶ UNOCHA, MSF, IMC, Insecurity Insight, etc. → Watchlist → GPIA Team

Georeferencing through Latitude and Longitude using:

- ▶ OpenStreetMap (OSM), Google Maps, bbox finder, news reports

Methodology

Data: attacks on healthcare



Methodology

Report No	Team Member	Report link	Report Org	Report date	Incident date	Duplicate Report?	Lat	Long	Precision	Notes	Detailed Description
006	Adi	http://reliefweb.int/sites/reliefweb.int/files/resources/a160324-acaps-south-sudan-conflict-food-insecurity.pdf	ACAPs	Incident February 17-18; Report Published 24 Mar 2016	15/02/16	-	7.709188	27.983474	town/city	-	Violence broke out between government forces and armed groups in Wau in Western Bahr el Ghazal. Houses have been burned and tens of thousands of people have been displaced to the bush.
014	Joëlle	http://www.insecurityinsight.org/ai-didanger/wp-content/uploads/2016/04/The-Aid-Security-Monthly-News-Brief-MAR-2016.pdf	Insecurity Insight	March 18, 2016	18/03/16	-	7.782388	33.002543	town/city	-	A Save the Children national employee working on a health program was shot dead in unclear circumstances.
016	Joëlle	http://reliefweb.int/sites/reliefweb.int/files/resources/160530_OCHA_SouthSudan_hb7.pdf	OCHA	May 30, 2016	15/05/16	-	4.095301	30.677401	town/city	-	Sister Veronika Rackeva, a Slovakian nun and medical doctor was shot while on a humanitarian mission, and later succumbed to her wounds. Sister Veronika was driving an ambulance on her way back from a medical center when she was attacked. Humanitarian partners have recently recorded an increase in attacks, robberies, and harassment against ambulances, particularly in Juba.
017	Joëlle	http://reliefweb.int/sites/reliefweb.int/files/resources/160509_OCHA_SouthSudan_humanitarian_bulletin.pdf	OCHA	May 2016 (precise incident dates unknown)	25/04/16	-	9.231487	29.800503	town/city	-	A rocket propelled grenade landed inside the perimeter of the compound and partly damaged one of the humanitarian hub accommodation containers.
019	Joëlle	http://reliefweb.int/sites/reliefweb.int/files/resources/160722_OCHA_SouthSudan_humanitarian_bulletin10.pdf	OCHA	July 22, 2016	07/07/16 - 10/07/16	-	4.819063	31.541381	town/city	-	Shelling/gunfire During the clashes, humanitarian facilities, compounds, and warehouses—including a maternity ward in a PoC site—were hit by shelling and gunfire, attacked or looted, and humanitarian staff were under attack
020	Joëlle	http://reliefweb.int/report/south-sudan/international-medical-corps-hospital-juba-hit-shelling-ambulance-along-violence	IMC	8 July, 2016	11/07/16	yes -	4.819063	31.536381	town/city	-	Shelling/gunfire The maternity wing of International Medical Corps' hospital in the Protection of Civilian (PoC) site in Juba, South Sudan, was hit by shelling today amidst escalating violence that is endangering the lives of thousands of people who fled into the camp for safety.

Reports → Points



Creation of New Layers



Interactive features

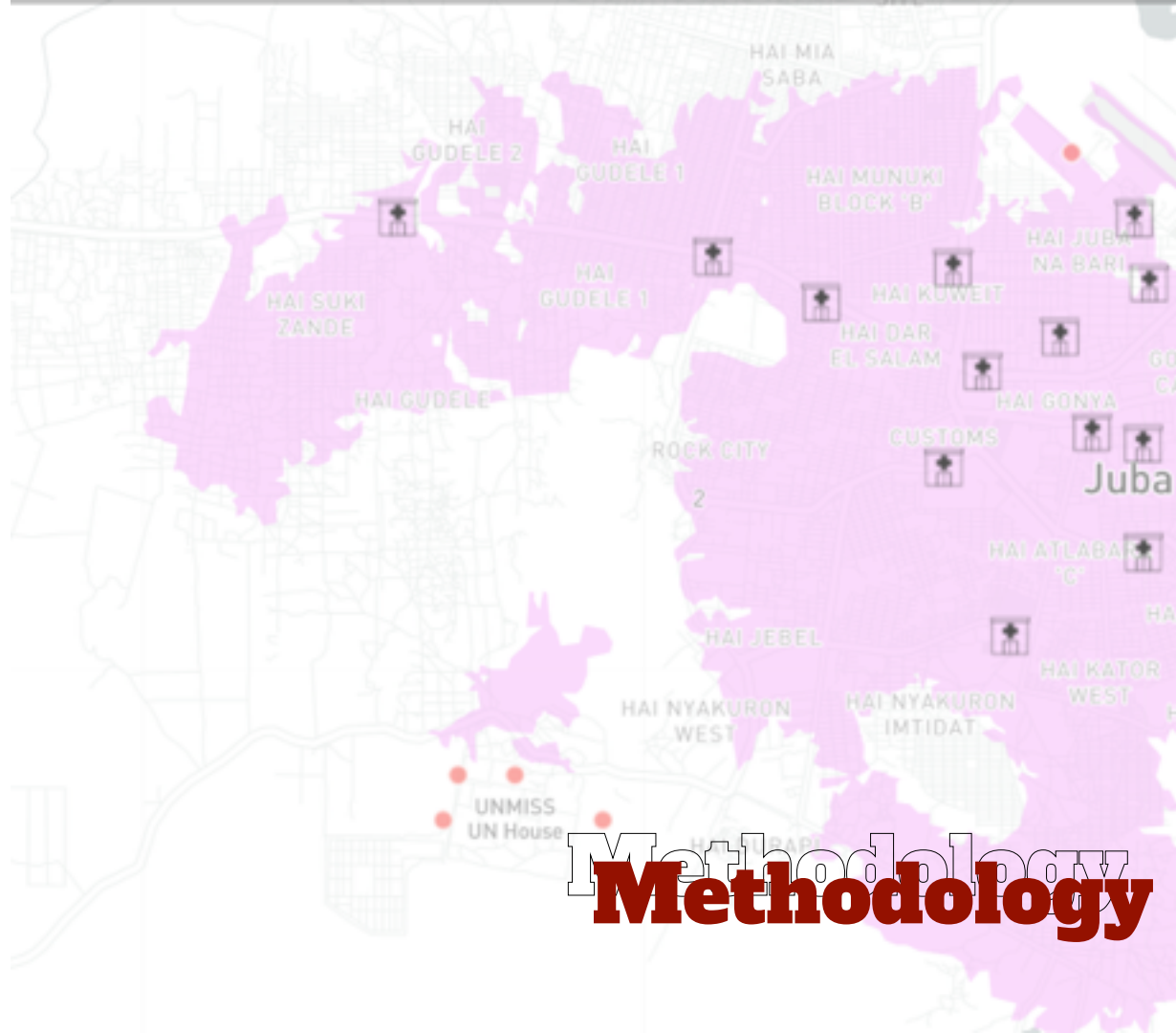
Methodology

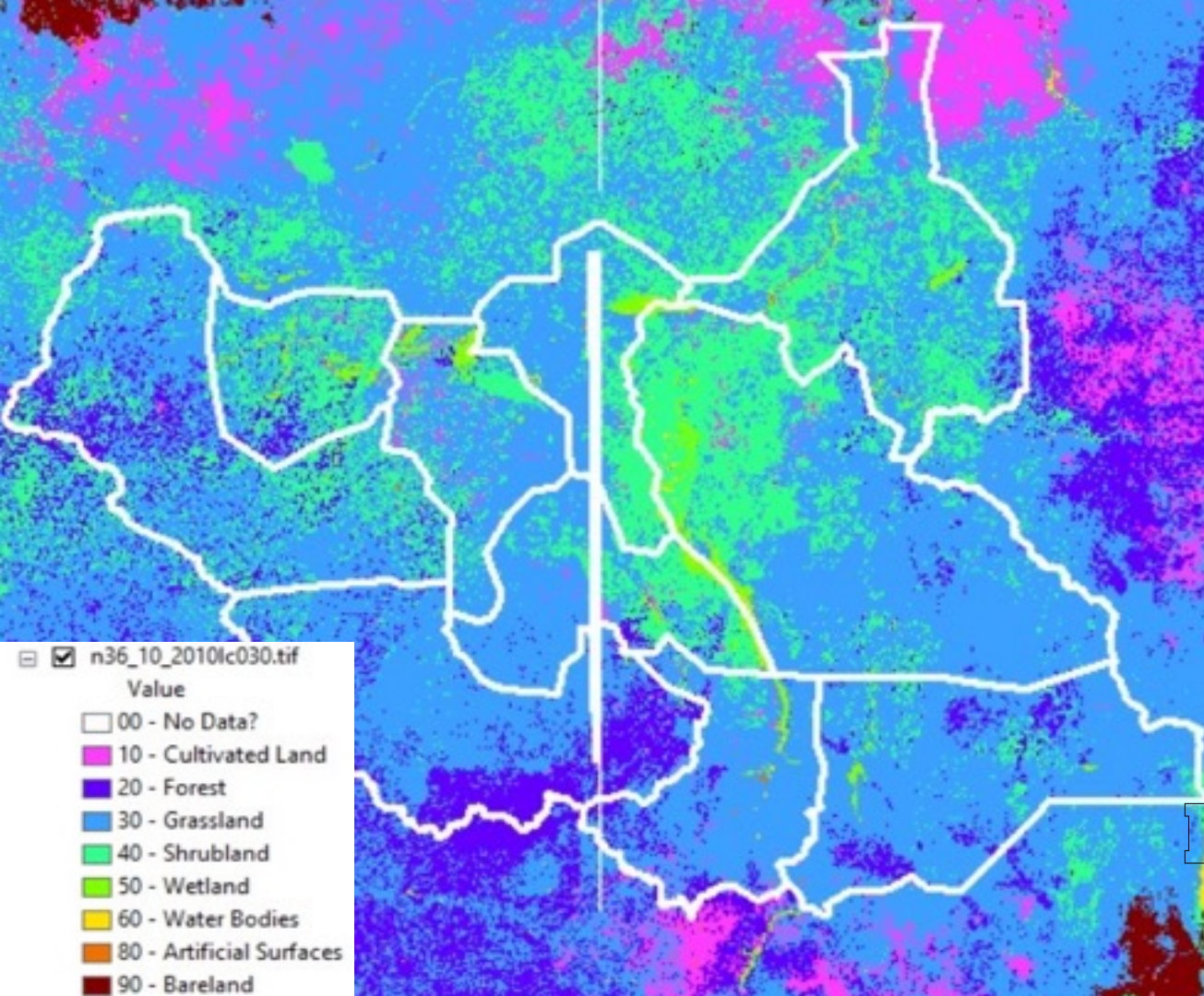
Step 2

Research and development of new layers

Additional layers include

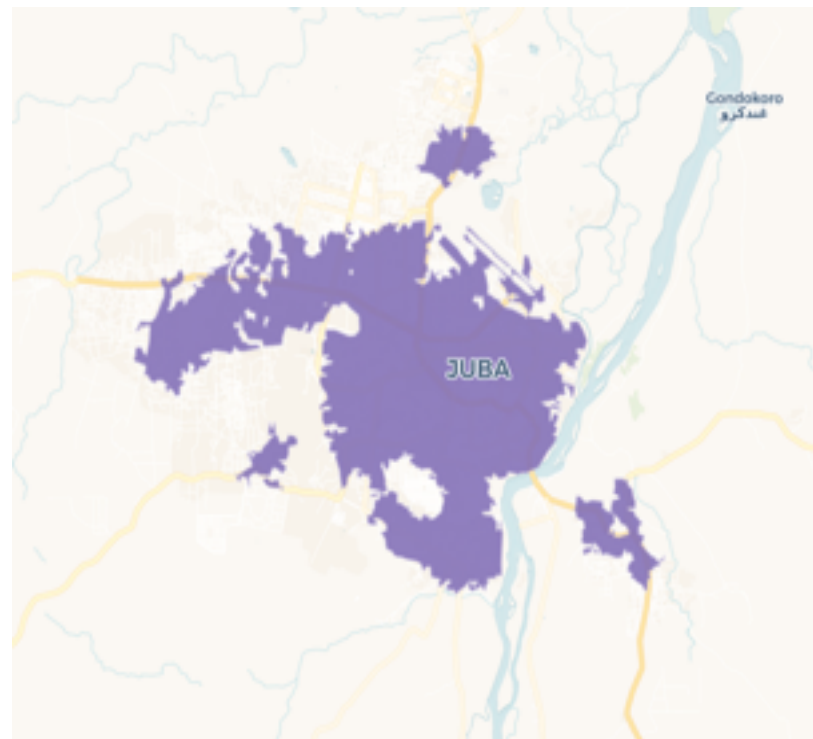
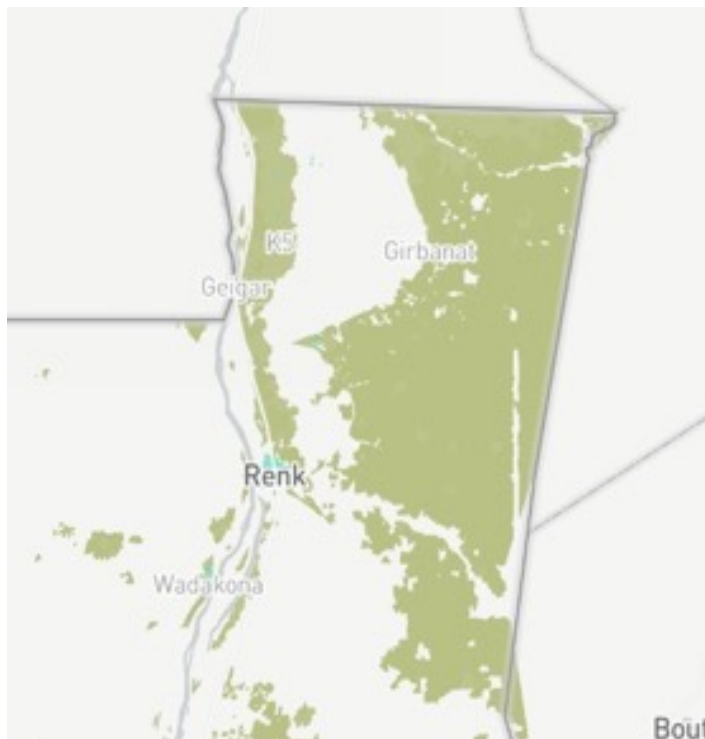
- Urban/Farmland
- IDP & Refugee camps
- Healthsites



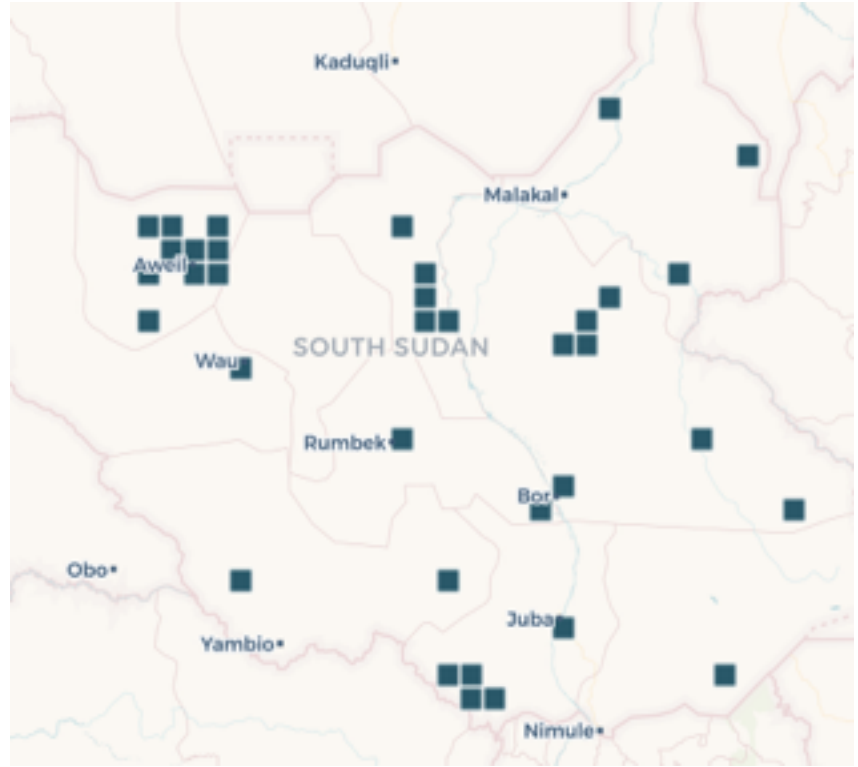
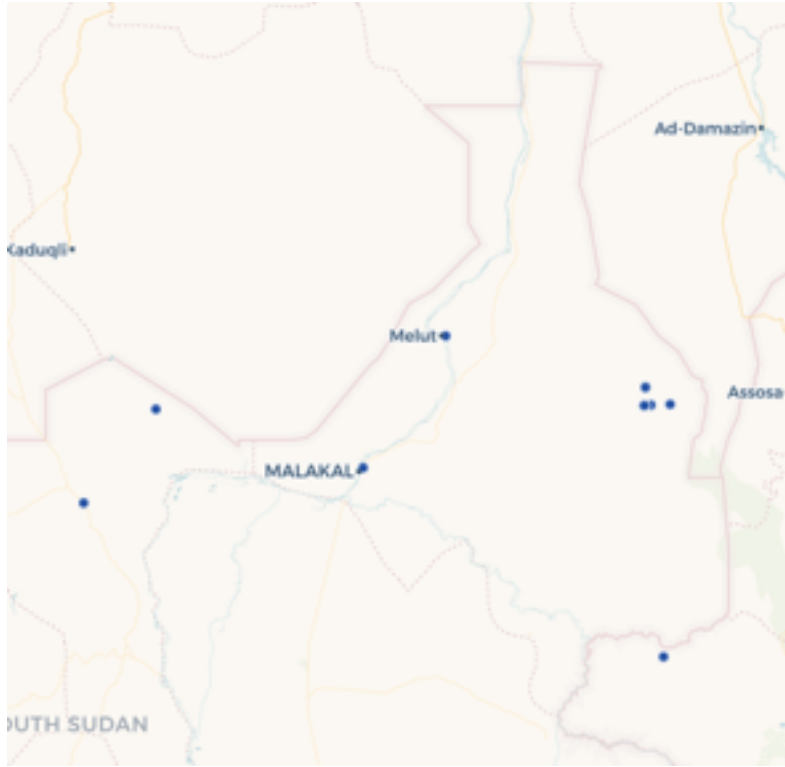


- ▶ Reclassified each individual land type with others as NoData.
- ▶ Converted new rasters into polygons (long, demanding process—30m product).
- ▶ Further developed selected polygons in Mapbox.

Methodology

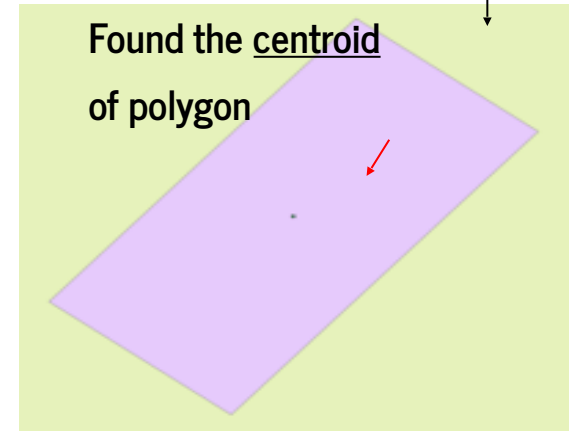
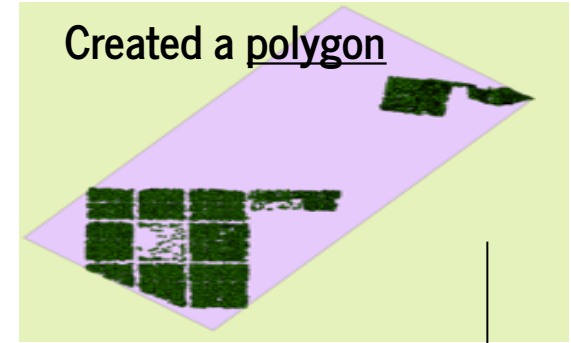
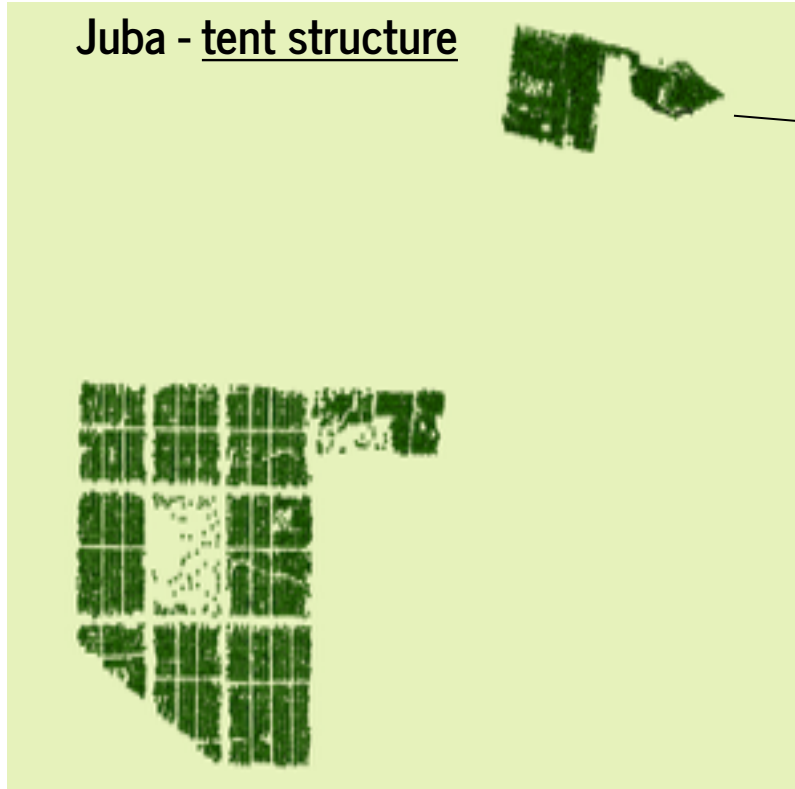


Research & development of new layers



Research & development of new layers

Tent
structure
to
Camp
Point
process:



Research & development of new layers

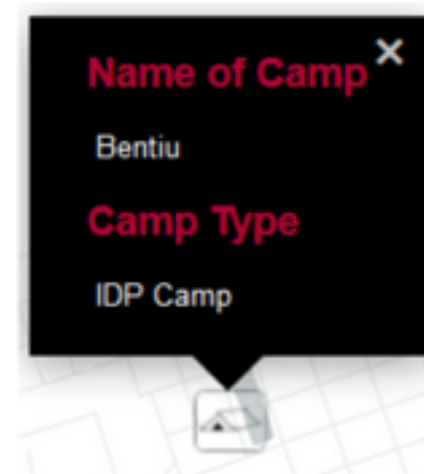
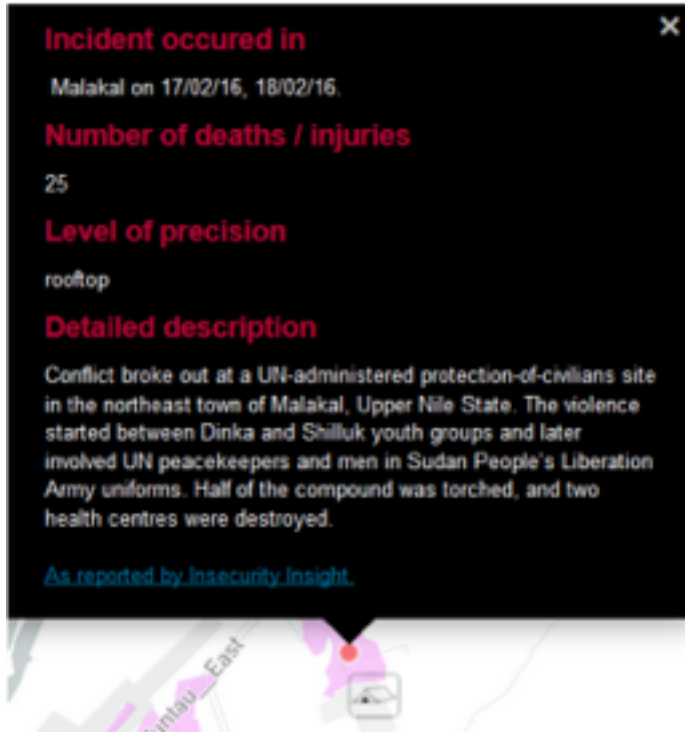
Step 3

Interactive features

- ▶ Popups
- ▶ Buttons
- ▶ Checkboxes

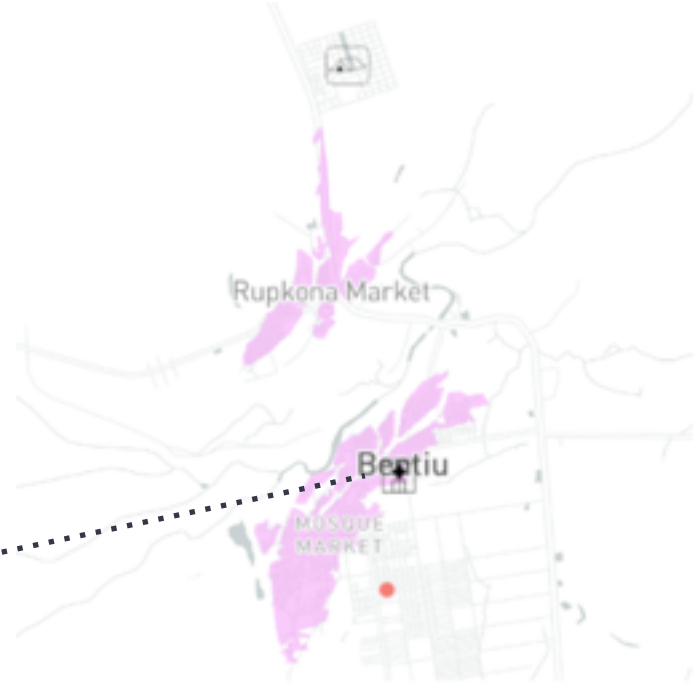


Pop up boxes



Interactive features

Zoom buttons



Interactive features

Button design + checkboxes

SOUTH SUDAN

JUBA

MALAKAL

WAU

BENTIU

☐  LOOTING

☐  SHELLING

☐ ○○○ OTHER

SHOW ALL PRECISION LEVELS 

Interactive features

Button design + checkboxes

SOUTH SUDAN

JUBA

MALAKAL

WAU

BENTIU

☒  LOOTING

☐  SHELLING

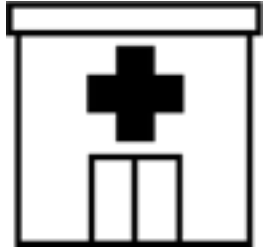
☒ ○○○ OTHER

SHOW ALL PRECISION LEVELS ⇅

Interactive features

```
Project
├── yemen
│   ├── index.html
│   ├── script.js
│   ├── styles.css
│   └── toadd
index.html
42 </label>
43 </div> -->
44
45 <!-- //// -->
46
47 <div class="boxes" style="padding-bottom:none;">
48 <div class="legend-title" style="font-family: Lato; font-we
49
50 <input type="checkbox" id="poly" class="poly-checkbox" che
51 <label for="poly">governorate-level</label>
52
53 <input type="checkbox" id="points" class="points-checkbox" cl
54 <label for="points">individual attacks</label>
55
56 <span style="padding-bottom:none;" class="cartodb-legend catego
57
58 <div class="legend-title" style="font-family: Lato; font-we
59
60 <ul class="legend-list" style="color:black;">
61
62 <li style="font-size:14px;font-family:Cardo;font-weight:90
63 <div class="bullet1" style="background-color:#e64d5a;"><
64 >75%
65 </li>
66
67 <li style="font-size:14px;font-family:Cardo;font-weight:bo
68 <div class="bullet1" style="background-color:#f97b57;"><
69 50% - 75%
70 </li>
71
248
249
250
251
252
253 // Update the SQL for the layer based on the current stat
254 function updateSql() {
255 var sql = "SELECT * FROM gilla549.revised_ssudan_incid
256 var whereConditions = [];
257 if ($('#looting-checkbox').prop('checked')) {
258 whereConditions.push("attack_type = 'Looting'");
259 }
260 if ($('#shelling-checkbox').prop('checked')) {
261 whereConditions.push("attack_type = 'Shelling'");
262 }
263 if ($('#shooting-checkbox').prop('checked')) {
264 whereConditions.push("attack_type = 'Shooting'");
265 }
266
267 // If there are any whereConditions, we add WHERE to the
268 if (whereConditions.length > 0) {
269 // Then we combine them with ' OR ', so any of the co
270 sql += ' WHERE ' + whereConditions.join(' OR ');
271 }
272 // Leaving this console.log() here so we can make sure
273 console.log(sql);
274 pointsLayer.setSQL(sql);
275 }
276
277
```

Front end design



Hospital/clinic icon



IDP/Humanitarian
site icon



Shelling

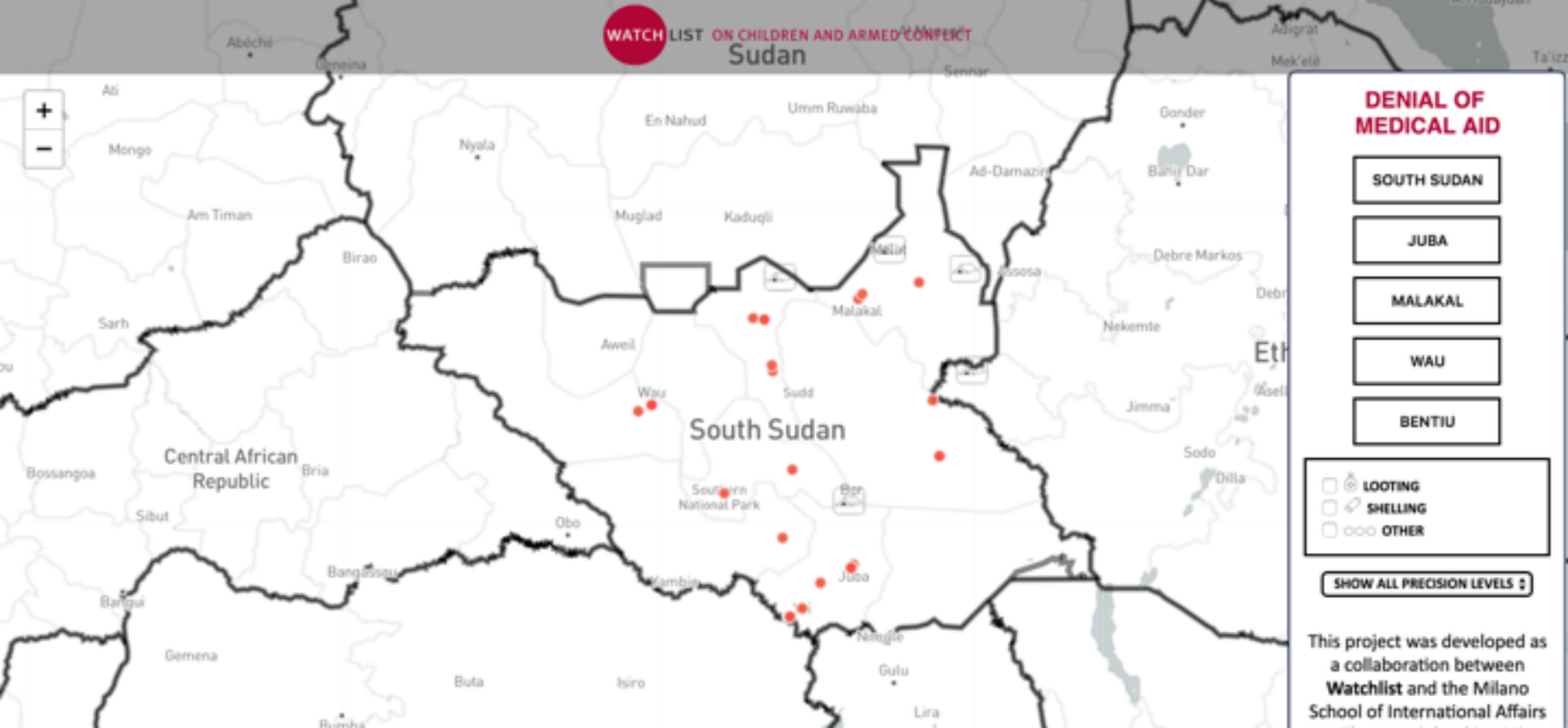


Looting



Other

Design elements



<https://dot-guarantee.glitch.me/>

Project status



Data collection  100%



Geo-referenced/Parsed data  100%



Back End  70%



Front End  70%

Project status

- Difficult to georeference map
- Changing topography and political map
- JavaScript steep learning curve
- Forced to simplify certain layers because of data size (farmland)
- Limited data
- Limitations of working with a client

Challenges
Challenges

- ▶ Deep understanding of conflict in South Sudan
- ▶ Acquired GIS and coding skills
- ▶ Enhanced design skills and expanded portfolio
- ▶ Gained experience working as a team and overcame challenges

Knowledge gained

Thank you
Thank you
Thank you